Understanding Generalization Requires Universal Induction

Aram Ebtekar
AIXI Labs
aramebtech@gmail.com

Marcus Hutter
Google DeepMind & ANU
www.hutter1.net

Danica J. Sutherland
UBC & Amii
dsuth@cs.ubc.ca

Abstract

Classical statistical theory is insufficient to explain the successes of general-purpose AI models, because it depends on handcrafted inductive biases that it cannot justify. No Free Lunch (NFL) theorems force any learner that beats chance on some environments to underperform on others. We might hope that past experience informs which environments to expect, but NFL applies equally to meta-learning. Thus, any method that makes meaningful predictions necessarily begins with an inductive bias external to the data. Choosing to bias toward short programs yields Solomonoff induction (SI), whose performance is competitive against all computable learners – albeit up to "constants" that become large when comparing against specialized methods that exploit background information. We therefore relativize SI to an *information vantage point*, biasing toward short programs with access to all preexisting information. This reframes the inductive bias: instead of seeking some absolute notion of simplicity, we favor *accessibility* with respect to our vantage point. An algorithm can only outpredict the relativized SI to the extent that its code contains additional information about the data, and no algorithm can generate such information. While SI is incomputable and hence not a practical algorithm, it provides a formal optimum for inference in the limit of infinite compute, and there is evidence to suggest that frontier AI systems roughly approximate it. Thus, the only known answer to meta-NFL is rooted in algorithmic information theory, which we should expect to play a fundamental role in explaining the generalization behavior of modern (and future) AI systems.

1 Introduction

Machine learning extrapolates from seen data to predict unseen data. Deductive logic alone does not force any relationship between the seen and unseen data; our *inductive* inferences necessarily depend on structural assumptions about the data, called *inductive bias*. That the Sun has risen each day so far does not logically imply that it will do so tomorrow,[1] but we would find life difficult to navigate without some confidence in tomorrow's sunrise. This is often justified via Occam's razor, preferring simple explanations – but what is a "simple" explanation, and why should we prefer one?

Theoretical analyses of learning problems often assert some form of "simplicity." To give an example, in one-dimensional density estimation, one can assume there is an underlying density function whose (say) third derivative is 1-Lipschitz, and then prove that an algorithm successfully estimates such densities [e.g. Tsy09, Section 1.2]. We might be satisfied to accept that assumption; it "feels sensible." As we apply machine learning to increasingly general problems, however, the required assumptions become more complex and less *a priori* defensible. To obtain a theoretical account of why learning works, one must argue for a particular choice of inductive bias.

[1] One day, it may engulf the Earth as a red giant, or it may shrink into a white dwarf as the Earth drifts away [EMD26]. Neither situation is *logically* necessary; it remains deductively possible that the rules of gravity will change tomorrow.

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This choice is commonly justified by pointing to earlier instances in which the proposed inductive bias was successful. In other words, a part of the learning algorithm is itself inductively meta-learned [Sch87; TP98; Hos+22] from a lifetime of experience; in turn, our lifetime learning algorithm is informed by billions of years of Darwinian evolution [HL22; Gol+26]. This suggests a promising case study: since it seems implausible that our microbial ancestors billions of years ago "knew" much, evolution may serve as a model of learning from a state of true ignorance.

Naive attempts to build a theory of learning from true ignorance, however, run into No Free Lunch theorems. In statistical learning terms, we have no *a priori* way to guarantee low generalization error. Using a model class for which uniform convergence holds guarantees that our algorithm does not substantially overfit (i.e. find a predictor that does much better on training data than on fresh data), but cannot guarantee that it does not underfit (i.e. have high training error). Whether these explanations of non-overfitting even apply to practical neural networks remains unclear after much investigation; there are meaningful negative results [NTS15; Zha+17; NK19; Jia+20; Gas+24], but also some positive indications [DR17; Zho+19; YBM21; Zho+22; Lot+24a; Lot+24b]. If they do apply, how do we know that practical neural networks will not underfit to a new problem? Alternatively, "soft-preference" techniques [see Wil25], such as structural risk minimization [VC74; Vap91] in a universal hypothesis class, can ensure that the model class will not underfit. The drawback is that they cannot guarantee non-overfitting unless the soft preference – the algorithm's inductive bias – happens to be suitable for the problem. It seems we can provably prevent overfitting or underfitting, but not both.

Despite No Free Lunch, it is an empirical fact that life and machines alike are often able to learn well. One might ask: why care about theory? Kawaguchi et al. [KKB22] identify three "practical roles" for generalization theory: to provide (1) guarantees on expected risk, (2) guarantees on the generalization gap, and (3) insights to guide model selection. Learning theory addresses role (2) for "small" hypothesis classes, showing that test error is near training error. (The use of a validation set can, in some ways, make any class effectively small [KKB22, Section 4].) As to roles (1) and (3), however, No Free Lunch theorems present extreme challenges. We can only say anything if we assume the function we are trying to learn lies in some "concept class." Why should it? This is both a philosophical challenge and a barrier for understanding what AI systems can – or should – do.

In Section 2, we expand on how No Free Lunch theorems preclude the possibility of learning without an inductive bias – and, in fact, a meta-inductive bias, which itself cannot be learned via experience nor evolution [Wol23]. Theoretical explanations of learning are possible by assuming structure *a priori*; however, today we see generalist AI models deployed on increasingly arbitrary tasks. What sort of inductive bias would account for their generalization performance on new problems?

With the advent of the attention mechanism [Vas+17] and reasoning modes [ElK+24; Guo+25; Sne+25], leading model classes are rapidly moving toward supporting general, even universal forms of computation [PBM21; MS24; Yan+25; LW26]. Solomonoff induction [Sol64] is a logical extreme in this direction, corresponding to Bayesian inference with a weighted mixture prior over all possible computer programs. Due to the halting problem [Tur37; Sip12, Section 4.2], Solomonoff induction is not computable, but it provides an idealization of learning in the limit of unbounded computation; it is a central object in algorithmic information theory [LV19; HQC24]. However, it necessarily chooses prior weights for each of the infinitely many possible computer programs. What justifies that prior?

Inspired by Schurz [Sch19], we argue in Section 3 that while there is no absolute winner among the class of all mathematically well-defined predictors, most of these predictors are inaccessible to us in practice. This motivates an inductive bias in favor of *accessibility* from a given *information vantage point*, which Section 4 formalizes as a universal computer equipped with an oracle. We prove that Solomonoff induction from this vantage point is optimal, in the sense that the only way to do better is to hardcode additional information about the dataset. Formally, Theorem 3 says that oracle Solomonoff induction's regret against any predictor is bounded by that predictor's (algorithmic) mutual information with the data – the extent to which it "knew the answer" in advance.

In Section 5, we discuss preliminary evidence supporting this point of view as a reasonable idealized description of modern AI systems. Appendices A and B cover additional details of oracle Solomonoff induction, Appendix C discusses what some key results in statistical learning theory (do not) say in relation to No Free Lunch, and Appendix D gives implications for the problem of Boltzmann brains in cosmology. Overall, we argue that **the only known answer to the meta-No Free Lunch problem is based on algorithmic information theory, and that this point of view gives a viable strategy for explaining the generalization behavior of modern (and future) AI systems**.

2 No Free Lunch in learning theory

No Free Lunch theorems have an extensive history, dating back to Hume's problem of induction [Hu1748; Put63; Goo83; Wol96; SL05; Ada+19; Bel20; SG21; Wol23]. Here, we focus on a straightforward formulation for finite input and output sets, whose proof is immediate by counting.

Proposition 1. *Consider a function $f^* : \mathbb{Z}_n \to \mathbb{Z}_k$, and let $D_{\text{train}} \subset \mathbb{Z}_n$ be a set of $|D_{\text{train}}| = m < n$ observations. Then, there are exactly k^{n-m} functions $f : \mathbb{Z}_n \to \mathbb{Z}_k$ consistent with these observations, in the sense that $f(x) = f^*(x)$ for all $x \in D_{\text{train}}$. At every unobserved $x' \in \mathbb{Z}_n \setminus D_{\text{train}}$, for every $y' \in \mathbb{Z}_k$, exactly k^{n-m-1} of the functions consistent with f^* on D_{train} have $f(x') = y'$.*

As a consequence of Proposition 1, no matter how a prediction algorithm uses the training dataset $(x, f^*(x))_{x \in D_{\text{train}}}$ to predict $f^*(x')$, it will fail for most possible target functions f^*.[2] To better appreciate the situation, let's examine how Bayesian and frequentist inference – representing two major schools of thought in statistical methodology – each fall short without an inductive bias.

Bayesian and frequentist no free lunches To avoid inductive bias, suppose the Bayesian statistician begins with a uniform prior over all k^n possible functions. After observing the training dataset $(x, f^*(x))_{x \in D_{\text{train}}}$, the posterior belief is a uniform distribution on the k^{n-m} functions consistent with the data. No generalization is achieved, because the belief on non-training points $x' \notin D_{\text{train}}$ remains uniform, assigning probability $k^{n-m-1}/k^{n-m} = 1/k$ to each possible value of $f^*(x')$.

Similarly, suppose the frequentist statistician samples a random hypothesis $\hat{f}$ uniformly[3] among the k^{n-m} functions that satisfy $\hat{f}(x) = f^*(x)$ for all $x \in D_{\text{train}}$, to test on a hold-out set $D_{\text{test}} \subsetneq \mathbb{Z}_n \setminus D_{\text{train}}$. Since $(\hat{f}(x))_{x \in D_{\text{test}}}$ is uniformly distributed, the test rejects $\hat{f}$ with overwhelming probability. We can follow up by sampling new hypotheses to test, but if all we do is sample at random until we obtain satisfactory performance on D_{test}, the resulting hypothesis will fail to generalize, remaining uniformly distributed on all $x' \notin D_{\text{train}} \cup D_{\text{test}}$.

We note an important difference between the two methodologies. The Bayesian statistician sets an explicit inductive bias at the start, via a prior. Thereafter, Bayes' rule determines optimal posterior inferences. This makes Bayesian methods readily automatable, perhaps contributing to their historical popularity in AI research. On the other hand, the frequentist statistician delivers an inductive bias in an online fashion, providing guidance in the form of hypotheses to test one after another. This is more amenable to experimental science: when human scientists are unable to encode their beliefs into an explicit prior, they benefit from the ability to make a sequence of guesses.

In these naive forms, neither methodology is self-contained: they are formally agnostic to the question of which inductive bias is most appropriate, outsourcing this choice to human design. With a good Bayesian prior or a good method of hypothesis generation, much better inferences become possible.

A concrete example Let $n = k = 19$, so that we want to learn a function $f^* : \mathbb{Z}_{19} \to \mathbb{Z}_{19}$. We observe $f^*(4) = 4$, $f^*(8) = 8$, and $f^*(11) = 11$. What is $f^*(0)$? Without further constraints, it can be anything: by Proposition 1, there are 19^{16} (about 288 quintillion) functions compatible with the data, with exactly 19^{15} (15 quintillion) of them assigning each possible value in $\mathbb{Z}_{19}$ to $f^*(0)$.

Nonetheless, it feels intuitive to guess the function $f_1(x) := x$, which predicts $f_1(0) = 0$. What makes f_1 a better guess than, say,

$$f_2(x) := x + (x-4)(x-8)(x-11) = x^3 - 4x^2 - 6x + 9 \pmod{19},$$

which also matches the data but predicts $f_2(0) = 9$? We might think of f_2 as representing a weaker inductive bias, in the sense of being derived by fitting the data to a large hypothesis class

$$\mathcal{H}_0 := \left\{ x \mapsto a_3 x^3 + a_2 x^2 + a_1 x + a_0 \pmod{19} : a_i \in \mathbb{Z}_{19} \right\}.$$

$\mathcal{H}_0$ contains $19^4 = 130{,}321$ elements, enough to *overfit* our small dataset. Hence, observing $f_2(x) = f^*(x)$ for $x \in D_{\text{train}} = \{4, 8, 11\}$ should not give us confidence that f_2 will continue to fit

[2] Note that this applies to all algorithms that depend only on the training data, regardless of how well they fit it, because such algorithms cannot distinguish between the k^{n-m} target functions that produce the same training labels.

[3] Alternatively, one could choose an "arbitrary" consistent hypothesis, which might perform better (or worse) than the random choice. Typical frequentist analyses therefore assume either a random choice, a *worst-case* choice (naturally only worse), or perhaps some particular tie-breaking rule (which would correspond to a choice of some non-uniform inductive bias).

future data. Since f_1 belongs to the much smaller hypothesis class

$$\mathcal{H}_1 := \{x \mapsto a_1 x + a_0 \pmod{19} : a_i \in \mathbb{Z}_{19}\},$$

of size only $19^2 = 361$, we expect it not to drastically overfit. The issue with this analysis is that f_2 is a member of not only $\mathcal{H}_0$, but also some smaller classes, such as the following, also of size 361:

$$\mathcal{H}_2 := \left\{x \mapsto x^3 - 4x^2 + a_1 x + a_0 \pmod{19} : a_i \in \mathbb{Z}_{19}\right\}.$$

Choosing $\mathcal{H}_2$ may feel like "cheating," since we constructed it by taking the leading terms $x^3 - 4x^2$ from f_2, which depended on the particular observations on D_{train}.[4] But, regardless of how we the authors selected it, what is it for you the reader that makes $\mathcal{H}_2$ "feel tailored" to the training data while $\mathcal{H}_1$ does not? If we remove our intuitive preference for f_1 and $\mathcal{H}_1$, the situation is formally symmetric. If the true function is f_1, then fitting to $\mathcal{H}_1$ performs well while fitting to $\mathcal{H}_2$ performs poorly; if the true function is f_2, the situation is reversed.

We visualize the space of possible functions in Figure 1. Learning theory [VC68; Val84; SB14; MRT18; Bac24], roughly, says that learning from a "small" hypothesis class (such as $\mathcal{H}_1$ or $\mathcal{H}_2$) probably yields hypotheses with a small generalization gap: their training performance is about the same as their test performance. Large classes, such as $\mathcal{H}_0$, lack this property: when using them, we may overfit the training data. Small classes are vulnerable to a different problem: if they contain no good approximation to the true function, then we *underfit*, i.e. the training performance is itself poor. Indeed, $\mathcal{H}_2$ underfits most random samples (of three or more points) labeled by f_1, and conversely $\mathcal{H}_1$ underfits most samples labeled by f_2.

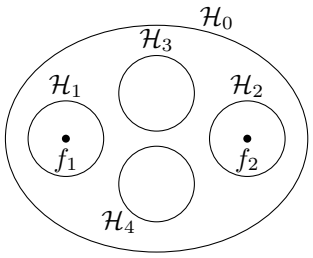


Figure 1: Both the "simple" function f_1 and the "complex" function f_2 belong to small hypothesis classes. The same can be said of any function in the large class $\mathcal{H}_0$. To generalize well, we should choose a class that is small while containing (a good approximation of) the true function.

Learning theory recommends choosing a small class to prevent overfitting, but does not tell us *which* small class prevents *underfitting*. No Free Lunch says that no algorithm generalizes well on most functions, so no hypothesis class simultaneously satisfies both constraints for most functions. We are forced to favor certain kinds of functions *a priori*. Why do we feel inclined to choose $\mathcal{H}_1$ over the myriad equally small classes such as $\mathcal{H}_2$? Low polynomial degree seems like a reasonable heuristic, but is not a general rule; it is disastrous for fitting other "simple" functions such as $f_3(x) := \lfloor x/10 \rfloor$.

Meta-learning an inductive bias Vitally, past experience cannot serve as a root justification for inductive biases: even assuming (generously) that we identify the full labeling function for every problem we encounter, meta-learning (learning to learn) is itself a learning problem [Wol23].

Corollary 2 (Meta-No Free Lunch)**.** *Take N prediction problems, with the ith revealing a function $g_i : \mathbb{Z}_n \to \mathbb{Z}_k$. After observing $g_1, \ldots, g_m$ with $m < N$, there remain $k^{(N-m)n}$ consistent meta-mappings $i \mapsto g_i$, exactly $k^{(N-m-1)n}$ of which predict each of the k^n possible choices for g_{m+1}.*

Proof. Uniquely label each function $g : \mathbb{Z}_n \to \mathbb{Z}_k$ by a number $\text{enc}(g) \in \mathbb{Z}_{k^n}$. Now consider the problem of learning the meta-function $f : \mathbb{Z}_N \to \mathbb{Z}_{k^n}$ given by $f(i) := \text{enc}(g_{i+1})$. The result follows from applying Proposition 1 with $n, k, D_{\text{train}}, x'$ replaced by $N, k^n, \mathbb{Z}_m, m$, respectively. □

As suggested in Section 1, let's consider how Nature might learn $f^* = f_1$ in a toy model of Darwinian evolution. Imagine a population of creatures, each genetically wired to compute some function $f : \mathbb{Z}_{19} \to \mathbb{Z}_{19}$. For simplicity, we assume an infinite population, providing enough initial variation so that no mutations are needed. On each day, the creatures are tested on a single input x; a creature f survives iff $f(x) = f_1(x) := x$. On the first three days, suppose the inputs are $4, 8, 11$. Then, all surviving creatures satisfy $f(4) = 4, f(8) = 8, f(11) = 11$. Do the creatures adapt under selection? Specifically, given a new input on day four, is the survival rate better than chance?

The answer depends on the initial population. This setting is formally identical to our Bayesian analysis above, with natural selection taking the role of Bayesian updates from a prior (an initial

[4] More extremely, we could pick (as do many students first exposed to learning theory) the singleton hypothesis class $\{f_2\}$.

population) to a posterior (a subsequent population). If the initial creatures are uniformly sampled from the set of all functions, then the generalization performance – the survival rate on day four – is only as good as chance. Without an explicit nonuniform bias at the outset, we can never learn one.

3 Accessibility as a universal inductive bias

In any realistic evolutionary setting, the initial distribution would not be uniform over all possible functions. In biology, perhaps even in the primordial soup where life began, molecules can arrange themselves into simple computing machines [Joh+22]. Thus, functions that can be computed by simple circuits, such as $f_1(x) := x$ or $f_3(x) := \lfloor x/10 \rfloor$, are sampled more frequently than others. This results in good generalization performance when testing on such functions.

The inductive bias toward simplicity is commonly known as Occam's razor, after the medieval philosopher William of Ockham. Its mathematical formalization as Solomonoff induction was proposed by Solomonoff [Sol64], and improved by Levin [Lev71]. Rathmanner and Hutter [RH11] advocate for it as a "gold standard" for induction, although this view is challenged by McGregor [McG14], Herrmann [Her20], Neth [Net23], and Sterkenburg [Ste26].

One major issue is that when simplicity is measured by description length, it depends on the choice of description language. Indeed, *any* prior distribution $\nu(x)$ can be interpreted as a simplicity bias, since a self-delimiting code of length $\lceil \log \frac{1}{\nu(x)} \rceil$ is shortest at the peaks of ν. Even Solomonoff's *universal* measures can be distributed arbitrarily on sequences of any prescribed finite length [LH15a]. Thus, "simplicity" appears to be a catch-all term for any possible preference we have.

Instead of seeking simplicity in absolute terms, Schurz [Sch19] argues that reality forces a meta-inductive bias toward inductive methods that are *accessible* to us. A method may be accessible because it is simple (e.g. "always predict `1`"), or because it is established (e.g. "run an online AutoML system to do evolutionary hyperparameter search over neural network architectures optimized by Adam, and use the predictor with the best cross-validation score"). A method may be *inaccessible* because it requires a specific random seed, or because it hardcodes unseen data (e.g. the test set, or next year's sporting results). We would like to ensure we don't do much worse than any methods we could plausibly use (i.e. have low regret against these predictors [CL06]); we are unbothered if our predictions are worse than those of a clairvoyant method we could never implement. We will see that a bias toward accessibility ensures predictions competitive with all accessible predictors; any predictor that does substantially better would necessarily be inaccessible to us.

Learning algorithms are themselves defined by (pseudo)code, which is a form of information. In general, we measure the inaccessibility of a piece of information (i.e. a string) by the length of the shortest "generalized pointer" (i.e. another string) to it. For example, some strings that point to the text of Shakespeare's *Hamlet* include, in order of decreasing length: (1) the full text verbatim; (2) the text in compressed `gzip` format, if the reader has e.g. `gzcat`; (3) the command "`curl -L flgr.sh/txtfssHamtxt`", if the reader has a Unix shell with Internet access; (4) the string "Text of Hamlet", if the reader has memorized the play or knows where to find a copy. Clearly, the accessibility of information is highly dependent on context. We refer to this context, representing the totality of information in our possession, as our *information vantage point*.

For concreteness, take the point of view of an AI program that lives inside a computer, interfacing with the outside world via a Bash shell. Its vantage point is defined by the connected storage and network devices (which include its own code), as well as the Bash interface. Together, these systems define a mapping from Bash commands to their outputs. As a result, some functions are accessible to the AI via short commands (e.g. simple calls to installed software libraries, or pretrained language models downloadable from HuggingFace), while others are not (e.g. someone else's private files, or next year's stock market movements).

The AI does not choose its vantage point; its (human or AI) designer does. Assuming a physical Church-Turing thesis, which states roughly that all physical processes are computable [Wol24], we can likewise model the designer as an algorithmic system with an exogenous information vantage point. To maximize the AI's predictive ability, the designer can grant the AI access to all of its information, some of which may happen to be usefully correlated with the unseen test data; however, the designer cannot *create* additional useful information (see Theorem 5).

Discussions of what the designer can or cannot do require some care: if the designer is itself an algorithm, then the outcome is predetermined [Lew81]. On the other hand, if we imagine counterfactually changing the AI program, then its predictions can be optimized by simply hardcoding the test data. To limit cheating by hardcoding, we take the middle ground of counterfactually optimizing only over very short programs. In order to model maximal intelligence, we run these short programs with unlimited time and space, enabling them to search for and retrieve larger codebases and datasets from the fixed information vantage point.

In fact, these short programs exactly capture the set of predictors that a designer can write with non-negligible probability, by randomly writing code like a monkey on a typewriter. One might object that it is also possible to sample some long programs with non-negligible probability, using either biased sampling, algorithmic search, or preexisting knowledge. However, all such cases are already covered by short programs at the fixed vantage point: biased or searchable samples can be compressed, and preexisting information can be downloaded with short commands.

The same reasoning applies more broadly than just programs running on silicon, such as in our model of Nature learning via Darwinian evolution. In the primordial soup, the most accessible functions are those which occur most frequently as molecules are stochastically rearranged [Deu13; Özk15; JW18; KW20; EH25]. As evolution selects for genetic information relevant to survival, future generations start from a richer information vantage point. In addition to this genetically hardwired information, we further enrich our vantage points through education, life experience, cultural context, and access to various repositories of information.

Section 4.2 will formalize the information vantage point as a universal computer U equipped with an oracle q, which can be queried for background information. If U represents a Bash shell on a computer, and q represents a snapshot of the local storage devices and World Wide Web, then a short program p can download a Python interpreter, packages, open-source repositories, and datasets, and adapt them with a custom Python script, to make a state-of-the-art predictor.[5] Thus, the central issue of which description language to adopt is resolved by granting access to all known languages via q.

In the spirit of No Free Lunch theorems, we should note that more data is not always better. The legendary Library of Babel [Bor41] is said to contain all possible 410-page books. This gigantic library is useless for the purposes of accessibility, since a set of directions pointing to one of its books would be as complex as the actual text (unless the library is designed to privilege certain books, e.g. placing *Hamlet* on an altar while leaving most other books on unremarkable shelves). There is a short program that prints the entire Library of Babel, including the book consisting of the first 410 pages of our test data, but that book by itself cannot necessarily be accessed by a short program.

4 Solomonoff induction

We now develop the formal theory of fully general optimal online learning with respect to an information vantage point. Our results are self-contained, deviating from standard presentations only where needed to make the vantage points explicit.

4.1 Bayesian mixture predictors

To balance simplicity with generality, we focus on optimizing the log loss (also called cross-entropy) in the setting of online binary sequence prediction. We make no distributional assumptions: the observation sequence $x \in \mathbb{B}^*$ ($\mathbb{B} := \{0, 1\}$) may be chosen by an adversary. To cover some common special cases, x may be formed by concatenating self-delimiting tokens, perhaps sampled i.i.d. from a continuously parametrized distribution [Hut04, Section 3.7.2].

[5] Strictly speaking, some of this code is unnecessary: the fact that AI researchers and engineers wrote it using mathematical reasoning and old datasets shows that it is in principle derivable from a poorer vantage point, such as one from the year 1960. The added contribution of our modern vantage point, which a learner from 1960 should not be expected to compete against, includes chaotic (unpredictable, even in principle) trends in arts, culture, and politics; as well as information from new sensing instruments, such as astronomical data from modern telescopes. The Transformer architecture is readily accessible from our vantage point. From the vantage point of 1960 or ancient history, it appears inaccessible; however, given infinite computation to search for good predictors, perhaps even Plato could have discovered the Transformer. It remains an open question how much of the architecture is a historical accident, and how much follows from the generic task of natural language prediction. Future work may benefit from studying time-constrained versions of accessibility; see [Gar+17; Xu+20; Fin+26; Voi+26].

We denote the length of x by $|x|$, its tth bit by x_t, and its first $t-1$ bits by $x_{<t}$. A *predictor* ν assigns next-bit probabilities $\nu(x_t \mid x_{<t}) \geq 0$ at each step t, forming the measure $\nu(x) := \prod_{t=1}^{|x|} \nu(x_t \mid x_{<t})$. If $\nu(\mathtt{0} \mid x) + \nu(\mathtt{1} \mid x) = 1$ for all $x \in \mathbb{B}^*$, then ν is a *probability measure*. To allow prediction by general programs that may eventually cease to output bits [LV19, Chapter 4; WH25], we only require that $\nu(\mathtt{0} \mid x) + \nu(\mathtt{1} \mid x) \leq 1$, and say ν is a *semimeasure* (short for *semiprobability measure*). The log loss of a predictor ν over the sequence $x \in \mathbb{B}^*$ is

$$L(\nu, x) := \log \frac{1}{\nu(x)} = -\sum_{t=1}^{|x|} \log \nu(x_t \mid x_{<t}). \tag{1}$$

Say we have a countable set of "expert" predictors $\{\nu_i\}_{i=1}^{\infty}$, with prior weights $w_i > 0$ (representing e.g. ν_i's prior credibility, or accessibility), $\sum_i w_i \leq 1$. We form the mixture-of-experts predictor

$$\xi(x) := \sum_{i=1}^{\infty} w_i \nu_i(x). \tag{2}$$

The mixture predictor ξ is competitive against (i.e. its loss is not much more than) every expert ν_i. Formally, we see this by using the fact that $\xi \geq w_i \nu_i$, to bound the *regret* of ξ against ν_i on x:

$$L(\xi, x) - L(\nu_i, x) = \log \frac{\nu_i(x)}{\xi(x)} \leq \log \frac{1}{w_i}.$$

This bound is independent of x, but depends on the weight w_i assigned to each expert. We might hope to make a universal mixture predictor by taking each possible program as an expert; unfortunately, the halting problem precludes such a mixture, or any good approximation, from being computable. Indeed, every computable predictor can suffer infinite regret against the predictor, computable in about the same time, that makes the opposite prediction with high confidence [Put63; Sol09, Section 3; Ste26]. We could compete against the latter by adding it to our mixture (2), but systematically adding many such experts drastically increases the computation time, while still allowing infinite regret against an analogous predictor. The joint optimization of time and sample efficiency presents a tradeoff [Sch02; Ven+11; FLH16; Nak21; Fin+26; Voi+26] whose analysis is beyond our scope.

Instead, we can already gain insights from idealizations [Wei07; GL08] that optimize sample efficiency in the infinite-time limit. There are several known constructions of limit-computable predictors that dominate all computable experts [Hut04, Section 2.4.3; HM07], all of which can be described as variants of "universal induction." They appear to have similar properties, so for simplicity we focus on the original Solomonoff-Levin choice of w_i and ν_i in (2) [Sol64; Lev71; Sol78]. To introduce it, we need some concepts from algorithmic information theory [LV19; HQC24].

4.2 Elements of algorithmic probability

Computers encode data and algorithms alike as binary strings. Given a finite string $x \in \mathbb{B}^*$, and a finite or infinite string $y \in \mathbb{B}^* \cup \mathbb{B}^\infty$, their concatenation $x_1 x_2 \ldots x_{|x|} y_1 y_2 \ldots$ is denoted xy, and we write $x \sqsubseteq xy$ to say that x is a prefix of xy. For $x, y \in \mathbb{B}^\infty$, their interleaving $x_1 y_1 x_2 y_2 \ldots$ is denoted $x \oplus y$. The infinite string of zeros is denoted $\mathbf{0} := \mathtt{0}^\infty$. Natural numbers $n \in \mathbb{N}$ are encoded as self-delimiting strings $\bar{n} \in \mathbb{B}^*$ with $|\bar{n}| = O(\log n)$, so that e.g. each pair $(n, x) \in \mathbb{N} \times \mathbb{B}^*$ is uniquely decodable from the string $\bar{n}x$.[6]

It will be convenient to fix a universal reference computer U, which we take to be a *monotone Turing machine* with four binary data tapes. It has a pair of read-only input tapes, one of whose contents we call an *oracle* and the other a *program*; a two-way read-write work tape; and a one-way write-only output tape.[7] We can think of the oracle tape as containing a downloaded snapshot of all available data, e.g. the World Wide Web. Alternatively, it may be more intuitive to view the oracle as a hardware abstraction layer that fetches data on demand.[8] For simplicity, we treat the data as fixed and ignore time and space complexity, making these two interpretations equivalent.

[6] One such construction: $\bar{n}$ consists of $\lfloor \log_2 n \rfloor$ 0s, followed by n's standard binary encoding.

[7] For readers unfamiliar with the Turing machine formalism, it will suffice to think of U as a computer that reads from two input streams and writes to one output stream. Once a bit is written, it can never be overwritten. The work tape functions as the working random-access memory (RAM). While many authors define the input tapes as one-way, this choice is not vital; one-way inputs can be copied to the work tape, or two-way input tapes can be prefixed with infinitely many zeros.

[8] Our oracle takes integer addresses instead of Bash commands containing URLs, but it is easy to translate between the two.

The tapes are infinitely long; this makes it possible, for example, to concatenate a randomized algorithm with its own infinite source of random bits, even though U is deterministic. Every cell of the work tape is initially set to 0. Given an oracle $q \in \mathbb{B}^\infty$ and program $p \in \mathbb{B}^\infty$, let $U^q(p) \in \mathbb{B}^* \cup \mathbb{B}^\infty$ denote the output of U, defined as the (finite or infinite) sequence of bits that it eventually writes. Since the output tape is one-way, this is well-defined regardless of whether U halts.

We choose U to be *universal* in the sense that there exists an effective enumeration of all such monotone Turing machines $\{T_i\}_{i \in \mathbb{N}}$, such that for all $i \in \mathbb{N}$ and $p, q \in \mathbb{B}^\infty$,

$$U^q(\bar{i}p) \;=\; T_i^q(p). \tag{3}$$

We can think of $\bar{i}$ as an interpreter for T_i in the language of U. Note that if q contains code for an interpreter, then we can choose T_i to be a computer that seeks and runs this interpreter from the oracle tape; this allows $\bar{i}$ to describe only a memory address instead of directly coding the interpreter. Consequently, fixing U is not a strong commitment to this particular computer; if we have access to additional computers, we can encode them as interpreters within q for easy access.

For a given program $p \in \mathbb{B}^\infty$, define the semimeasure *programmed* by p as

$$\mu_p(x) \;:=\; \mathbb{P}_{\alpha \sim \lambda}\left(x \sqsubseteq U^\alpha(p)\right). \tag{4}$$

λ denotes the Lebesgue measure on $\mathbb{B}^\infty$, meaning the bits of $\alpha \in \mathbb{B}^\infty$ are independent fair coin flips. $\mu_p(x)$ gives the fraction of random oracle tapes α for which the program p's output starts with x. Using (3), the program $\bar{i}\mathbf{0}$ corresponds to the semimeasure $\mu_{\bar{i}\mathbf{0}}(x) = \mathbb{P}_{\alpha \sim \lambda}(x \sqsubseteq T_i^\alpha(\mathbf{0}))$. Hence, $\{\mu_{\bar{i}\mathbf{0}} : i \in \mathbb{N}\}$ is the set of all lower-semicomputable semimeasures [LV19, Theorem 4.5.2], which properly includes the set of all computationally feasible predictors.

We also define the Solomonoff-Levin semimeasure relative to a particular oracle $q \in \mathbb{B}^\infty$ by

$$M^q(x) \;:=\; \mathbb{P}_{\alpha \sim \lambda}\left(x \sqsubseteq U^q(\alpha)\right), \tag{5}$$

the fraction of random programs α whose output starts with x when given *access* to q. M^q has a positive probability of sampling from every machine T_i, and can therefore be viewed as a universal mixture [HQC24, Theorem 3.8.8]. Concretely, since $\mathbb{P}_{\alpha \sim \lambda}(\bar{i} \sqsubseteq \alpha) = 2^{-|\bar{i}|}$, M^q equals the mixture (2) with $w_i := 2^{-|\bar{i}|}$ and $\nu_i(x) := \mathbb{P}_{\alpha \sim \lambda}(x \sqsubseteq T_i^q(\alpha))$. There exists a fixed machine T_j (independent of p, q) satisfying $T_j^q(p) = U^p(q)$, so that in particular $\nu_j = \mu_q$. Hence, using (3), (4), and (5), $\mu_{\bar{j}q} = M^q \geq w_j \nu_j = 2^{-|\bar{j}|}\mu_q$. Since j is fixed, we can write more compactly

$$M^q(x) \overset{\times}{\geq} \mu_q(x), \tag{6}$$

where the relational operators $\overset{\times}{\leq}, \overset{\times}{\geq}$ hide a multiplicative factor that depends only on U (and $\overset{\times}{=}$ means that both $\overset{\times}{\leq}$ and $\overset{\times}{\geq}$ hold). Similarly, we use $\overset{+}{\leq}, \overset{+}{\geq}, \overset{\pm}{=}$ to hide an additive term that depends only on U. Since the hidden constants are universal, independent of specific learning algorithms or data, U can be chosen in such a way as to make all the constants in this paper small [LV19, Section 3.9].

We can think of the information vantage point U^q as a specialized computer, obtained by equipping the fixed reference computer U with observer-dependent data q. Classical Solomonoff induction predicts according to the Bayesian prior $M^{\mathbf{0}}$, which may suffer high regret against a specialized expert that takes background information into account. By placing the observer's information in q, Theorem 3 will show that *oracle Solomonoff induction* using M^q has low regret against any expert that is accessible to us in practice. In particular, we need not worry about using the "wrong" universal computer: if we have access to a more suitable computer, it would be encoded within q.

4.3 Sample-optimal online learning with oracle Solomonoff induction

Shannon information theory studies the entropy and mutual information of random variables. It is readily applied to settings that sample repeatedly from i.i.d. or stationary ergodic sources, for which aggregate codelengths concentrate near their expectations [CT06, Chapter 3 and 16.8].

Algorithmic information theory defines analogous quantities for individual samples without associated distributions [GV04; LV19]. Some argue this makes algorithmic information theory relevant to more general (e.g. non-ergodic) settings [Kol83; EH25, Section IV.B]. For our analysis, a useful analogue to the Shannon conditional entropy $H(X \mid Q)$ (for random variables Q, X) is the *a priori complexity* of $x \in \mathbb{B}^* \cup \mathbb{B}^\infty$ relative to $q \in \mathbb{B}^\infty$, given by

$$KM^q(x) \;:=\; \log \frac{1}{M^q(x)}. \tag{7}$$

The logarithm's base amounts to a choice of units [Fra05]. If taken in base 2, KM^q is expressed in bits, and is typically slightly less than the prefix Kolmogorov complexity [HQC24, Theorem 3.8.7].

In Shannon theory, the conditional mutual information is $I(P; X \mid Q) := H(X \mid Q) - H(X \mid P, Q)$ for random variables P, Q, X. Analogously, we define an asymmetric notion of algorithmic mutual information between $p \in \mathbb{B}^\infty$ and $x \in \mathbb{B}^*$, relative to $q \in \mathbb{B}^\infty$, by

$$MI^q(p : x) \;:=\; KM^q(x) - KM^{p \oplus q}(x) \;=\; \log \frac{M^{p \oplus q}(x)}{M^q(x)}. \tag{8}$$

Recall that $p \oplus q$ is the interleaving of p and q, providing oracle access to both strings. Other definitions of algorithmic mutual information appear in the literature [Gác21, Section 3.1; Ver21], but MI^q is simple and useful for our purposes. Appendix A establishes some intuitive properties: Proposition 4 shows that $0 \stackrel{+}{\leq} MI^q(p : x) \stackrel{+}{\leq} \min(KM^q(p), KM^q(x))$, and Theorem 5 proves that information cannot be created *ex nihilo* by any small program, even with randomization.

The following bound states that if a predictor μ_p substantially outperforms M^q, its code p must contain additional information about x that is not in q. Since information cannot be created, there is no way to invent such a predictor. Thus, M^q beats the sample-efficiency of all realistic predictors.

Theorem 3 (Regret of oracle Solomonoff induction)**.** *For all $p, q \in \mathbb{B}^\infty$, the regret of a universal predictor M^q against another predictor μ_p, on any data $x \in \mathbb{B}^*$, is bounded by*

$$L(M^q, x) - L(\mu_p, x) \;\stackrel{+}{\leq}\; MI^q(p : x) - MI^p(q : x) \;\stackrel{+}{\leq}\; MI^q(p : x).$$

Proof. Using Equations (1) and (6),

$$L(M^q, x) - L(\mu_p, x) = \log \frac{\mu_p(x)}{M^q(x)} \stackrel{+}{\leq} \log \frac{M^p(x)}{M^q(x)} = \log \frac{M^{p \oplus q}(x)}{M^q(x)} - \log \frac{M^{p \oplus q}(x)}{M^p(x)}.$$

Using the definition (8), the first term is $MI^q(p : x)$. Since $M^{p \oplus q}(x) \stackrel{\times}{=} M^{q \oplus p}(x)$, the second term is $\stackrel{+}{=} -MI^p(q : x)$, which is $\stackrel{+}{\leq} 0$ by Proposition 4. □

Combining Theorem 3 and Proposition 4 yields the weaker bound $L(M^q, x) - L(\mu_p, x) \stackrel{+}{\leq} KM^q(p)$, which generalizes Corollary 3.8.10 of [HQC24]. Theorem 3 goes further by clarifying what type of information is relevant to the regret. For example, suppose $p = r\alpha$ consists of source code r for a randomized algorithm, along with a typical sample α from an infinite source of random bits, neither of which is tailored to the data x. Then $KM^q(p)$ is infinite, while $MI^q(p : x)$ is small.

Although oracle Solomonoff induction is formally Bayesian, Theorem 3 provides a distribution-free frequentist guarantee. This suggests an *interpretation* of probabilities [Gil00; HH16, Part IV]: no matter what x the world presents, an idealized learner may model its subjective uncertainty using M^q [RH11]. Our position does not take a side in the Bayesian-frequentist divide [May18; SH19], but may help illuminate why it exists. Given the framing from Section 2, where Bayesian inference fixes its inductive bias up front and frequentist inference accumulates it online, an idealized learner with unlimited resources can afford to commit to a universal prior, and is naturally Bayesian. A resource-bounded learner cannot, so its inductive bias must accumulate over time. Any realistic attempt to approximate M^q would find a never-ending sequence of candidate programs; therefore, we can expect practical methods to acquire a mix of Bayesian and frequentist character [SdH22].

5 Neural network generalization

Some of the most famous results in deep learning are "universal approximation theorems," which say that neural networks can approximate arbitrary functions [Cyb89; Hor91; KL20]. Referring back to Figure 1, this is like saying neural networks are a large class; they will fit the training data, but uniform convergence will not apply. While this is frequently cited as "the reason" that neural networks work well in practice, this is misleading: equivalent universal approximation results are shared by Gaussian kernel spaces [SC08, Section 4.6] and even histograms [RF10, Section 7.4].

In order to generalize well, we need more: a universal inductive bias, which approaches something like M^q in a limit of large model size and compute. The information vantage point U^q may include training data; before seeing data, it may be informed by common computing primitives. There are

some indications that modern models may correspond to a simplicity bias in universal computing terms [VCL19; Min+25]. Such arguments have recently been made to explain various empirical results in practical networks [Del+24; Gol+24; Huh+24; Hua+24], as well as in toy models [RS24].

How does this bias occur? In addition to mechanisms discussed by the previous papers, Buzaglo et al. [Buz+24] prove that wide feedforward networks have disproportionately large regions of parameter space corresponding to functions also computed by much narrower networks. Thus, a random search for interpolating solutions is more likely to correspond to a narrow network, supporting empirical findings that random search generalizes well [Chi+23; PKB26]. Narrow networks are not quite the same as short programs, but they are closely related, as small feedforward networks can directly encode small binary circuits [Par94]. Adding recurrence and an external memory turns circuits into universal Turing machines; transformers with chain of thought move toward that [PBM21; MS24; YW24; Yan+25; LW26]. Whether the parameter space bias extends to favoring short programs, and the consequences for practical optimization methods, remain to be seen.

Jacot [Jac26] suggests another distinct mechanism: under certain assumptions, a particular norm of ResNet parameters is closely related to the size of a minimal circuit implementing the same function. It is not yet clear whether practical optimization algorithms minimize that parameter norm.

Large language models learn to learn: after pre-training, they are able to "learn in-context" within a text sequence [Don+24], in a manner that has been argued to resemble Bayesian inference [Xie+22; WS26], or even specifically Solomonoff induction [WM25]. Without needing any real data at all, Grau-Moya et al. [Gra+24] train a neural network to approximate Solomonoff induction in-context, using synthetic data generated by random programs with a time bound. Cowsik et al. [Cow+26] extend this approach by co-evolving a generator for the random programs. A related line of work by Shaw et al. [Sha+26] trains a neural network to approximate the Minimum Description Length principle [BL03; Wal05; Hut09; GR19; ZS25; Li+26].

According to No Free Lunch, fitting to training inputs, even if sampled from a Solomonoff measure, still allows for arbitrary behavior on unseen inputs. Therefore, the strong generalization performance that these authors elicit on unseen inputs cannot be solely attributed to their synthetic data. We should investigate the extent to which successful machine learning algorithms and architectures exhibit a pseudo-universal inductive bias, prior to taking any training data.

6 Discussion

Goldblum et al. [Gol+24] recently argued that algorithmic information theory explains how to beat No Free Lunch, because "low-complexity structure shared by real-world datasets and machine learning models enables broad generalization across domains and sample sizes with a single model class." We do not disagree, but we ask: low complexity in terms of *what*, and *why* [Ste26]? Any argument based on empirical success – as theirs is – reverts back to the problem of meta-induction as in Corollary 2. Our accessibility framing provides a crisper account of optimal learning: while we cannot be sure that M^q will predict accurately, Theorem 3 guarantees that no realistic alternative is substantially better.

This framing brings prescriptive value: generalist AI systems can aim to approximate the regret bound in Theorem 3, at least when their compute budget greatly exceeds that of an expert μ_p. It is also of descriptive value: there is reason to think that deep learning and transformers have moved us closer to universal induction. We hope to apply this perspective toward understanding what AI systems do, and what they *should* do in challenging settings, such as in-context out-of-distribution generalization. Where real AI systems fall short of true universality – due to computational bounds or other differences – we expect M^q to serve as a useful baseline from which to understand deviations. Models of agency based on M^q, such as AIXI, can help us anticipate future agentic capabilities and behaviors, and identify safety risks before they occur [HQC24, Chapter 15; Meu+25; EC26].

Furthermore, we saw that the study of universal induction inspires additional concepts such as the algorithmic mutual information, which quantifies a model's knowledge of a dataset. Algorithmic information theory is a mathematical discipline replete with concepts relevant to machine learning, accounting for factors such as accessibility, model size, time bounds, and even novelty [FLH16; VS16; LV19; Zaf25; EWJ25; Fin+26; EC26; Voi+26]. We believe its applications are severely underexplored, and its relevance grows rapidly with advancing AI capabilities.

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A Additional properties of algorithmic mutual information

Throughout this section, it will be convenient to take KM and MI to be measured in bits (i.e. using base 2 logarithms in (7) and (8)), which are also the length unit for bit strings.

A.1 Supporting derivations

Our first result is analogous to the Shannon theory inequality for random variables P, Q, X:

$$0 \le I(P; X \mid Q) \le \min\left(H(P \mid Q),\, H(X \mid Q)\right).$$

Proposition 4 (Information bounds)**.** *For all $p, q \in \mathbb{B}^\infty$ and $x \in \mathbb{B}^*$,*

$$0 \le KM^q(x) \overset{+}{\le} |x|, \qquad 0 \overset{+}{\le} MI^q(p:x) \overset{+}{\le} \min(KM^q(p),\, KM^q(x)).$$

Proof. By considering the machine $T_k^q(\alpha) := \alpha$ that copies its program to the output tape, we obtain

$$\begin{aligned} M^q(x) &= \mathbb{P}_{\gamma\sim\lambda}\left(x \sqsubseteq U^q(\gamma)\right) \\ &\ge 2^{-|\bar{k}|-|x|}\mathbb{P}_{\alpha\sim\lambda}\left(x \sqsubseteq U^q(\bar{k}x\alpha)\right) \\ &= 2^{-|\bar{k}|-|x|}\mathbb{P}_{\alpha\sim\lambda}\left(x \sqsubseteq T_k^q(x\alpha)\right) \\ &= 2^{-|\bar{k}|-|x|}\mathbb{P}_{\alpha\sim\lambda}\left(x \sqsubseteq x\alpha\right) \\ &= 2^{-|\bar{k}|-|x|}. \end{aligned}$$

Taking logarithms and applying (7) yields $0 \le KM^q(x) \le |\bar{k}| + |x| \overset{+}{\le} |x|$.

By considering the machine $T_i^{p\oplus q}(\alpha) := U^q(\alpha)$ that ignores p, we obtain $M^{p\oplus q}(x) \ge 2^{-|\bar{i}|}M^q(x) \overset{\times}{=} M^q(x)$. Hence from the definition (8), $0 \overset{+}{\le} MI^q(p:x) \le KM^q(x)$.

For the remaining inequality, consider the machine $T_j^q(\alpha \oplus \beta) := U^{U^q(\beta)\oplus q}(\alpha)$. This involves an abuse of notation, as $U^q(\beta)$ is not necessarily an infinite string; if the machine tries to access an invalid index, we take it to cease writing to the output tape. Since j is a universal constant,

$$\begin{aligned} M^q(x) &= \mathbb{P}_{\gamma\sim\lambda}\left(x \sqsubseteq U^q(\gamma)\right) \\ &\overset{\times}{\ge} \mathbb{P}_{\alpha,\beta\sim\lambda}\left(x \sqsubseteq U^q(\bar{j}(\alpha\oplus\beta))\right) \\ &= \mathbb{P}_{\alpha,\beta\sim\lambda}\left(x \sqsubseteq T_j^q(\alpha\oplus\beta)\right) \\ &= \mathbb{P}_{\alpha,\beta\sim\lambda}\left(x \sqsubseteq U^{U^q(\beta)\oplus q}(\alpha)\right) \\ &\ge \mathbb{P}_{\beta\sim\lambda}(U^q(\beta) = p)\cdot\mathbb{P}_{\alpha\sim\lambda}\left(x \sqsubseteq U^{p\oplus q}(\alpha)\right) \\ &= M^q(p)\cdot M^{p\oplus q}(x). \end{aligned}$$

By rearranging and taking logarithms, $MI^q(p:x) \overset{+}{\le} KM^q(p)$. □

Next, we show that no deterministic or randomized meta-algorithm r can reliably create p that has high mutual information with x. Levin [Lev84] and Vereshchagin [Ver21] proved similar results for symmetric notions of mutual information, under the name *conservation of randomness*; Ebtekar and Hutter [EH25] frame one such result as a generalization of the second law of thermodynamics.

Theorem 5 (No information ex nihilo)**.** *Let $r \in \mathbb{B}^*$ be such that the randomly generated string $p := U^q(r\beta)$ is infinitely long for almost all $\beta \sim \lambda$. Then, for all $x \in \mathbb{B}^*$,*

$$\mathbb{E}_p\left[MI^q(p:x)\right] \le \log_2 \mathbb{E}_p\left[2^{MI^q(p:x)}\right] \overset{+}{\le} |r|.$$

Proof. Consider again the machine $T_j^q(\alpha \oplus \beta) := U^{U^q(\beta)\oplus q}(\alpha)$. A random string $\gamma \sim \lambda$ has probability $2^{-|\bar{j}|-|r|}$ of having the form $\bar{j}(\alpha \oplus r\beta)$ for some $\alpha, \beta \in \mathbb{B}^\infty$. Since j is constant,

$$\begin{aligned}
M^q(x) &= \mathbb{P}_{\gamma\sim\lambda}\left(x \sqsubseteq U^q(\gamma)\right) \\
&\overset{\times}{\geq} 2^{-|r|}\mathbb{P}_{\alpha,\beta\sim\lambda}\left(x \sqsubseteq U^q(\bar{j}(\alpha \oplus r\beta))\right) \\
&= 2^{-|r|}\,\mathbb{E}_{\beta\sim\lambda}\,\mathbb{P}_{\alpha\sim\lambda}\left(x \sqsubseteq T_j^q(\alpha \oplus r\beta)\right) \\
&= 2^{-|r|}\,\mathbb{E}_{\beta\sim\lambda}\,\mathbb{P}_{\alpha\sim\lambda}\left(x \sqsubseteq U^{U^q(r\beta)\oplus q}(\alpha)\right) \\
&= 2^{-|r|}\,\mathbb{E}_{p}\,\mathbb{P}_{\alpha\sim\lambda}\left(x \sqsubseteq U^{p\oplus q}(\alpha)\right) \\
&= 2^{-|r|}\,\mathbb{E}_{p}\,M^{p\oplus q}(x).
\end{aligned}$$

Rearranging, and applying the definition (8),

$$\mathbb{E}_p\left[2^{MI^q(p:x)}\right] = \frac{\mathbb{E}_p\,M^{p\oplus q}(x)}{M^q(x)} \overset{\times}{\leq} 2^{|r|}.$$

The result now follows from Jensen's inequality. □

Suppose we want to design a predictor μ_p that substantially outperforms M^q. According to Theorem 3, this requires $MI^q(p:x)$ to be high. What kind of process can generate a suitable program p? If we accept the physical Church-Turing thesis, the design process must be controlled by a preexisting meta-algorithm, perhaps inside our brain, making it accessible from our information vantage point. We can model any accessible method as using very little new code r, which uses the information vantage point q and random source β to generate $p := U^q(r\beta)$. Theorem 5 says that such a process almost always produces low $MI^q(p:x)$, so μ_p will not predict substantially better than M^q.

For the purposes of the following subsection, we need one additional result.

Theorem 6 (Information supplied by a self-delimiting prefix)**.** *Suppose $p, q \in \mathbb{B}^\infty$ and $r, x \in \mathbb{B}^*$, with r belonging to a fixed computably enumerable set, no member of which is a proper prefix of another. Then,*

$$MI^q(rp:x) \overset{+}{\leq} |r| + MI^q(p:x).$$

Proof. Since r is self-delimiting, there exists a fixed machine T_j that reads r, and simulates U on a modified oracle and program, so that $T_j^{p\oplus q}(r\alpha) := U^{(rp)\oplus q}(\alpha)$. A random string $\gamma \sim \lambda$ has probability $2^{-|\bar{j}|-|r|}$ of starting with $\bar{j}r$. Since j is constant,

$$\begin{aligned}
M^{p\oplus q}(x) &= \mathbb{P}_{\gamma\sim\lambda}\left(x \sqsubseteq U^{p\oplus q}(\gamma)\right) \\
&\overset{\times}{\geq} 2^{-|r|}\mathbb{P}_{\alpha\sim\lambda}\left(x \sqsubseteq U^{p\oplus q}(\bar{j}r\alpha)\right) \\
&= 2^{-|r|}\mathbb{P}_{\alpha\sim\lambda}\left(x \sqsubseteq T_j^{p\oplus q}(r\alpha)\right) \\
&= 2^{-|r|}\mathbb{P}_{\alpha\sim\lambda}\left(x \sqsubseteq U^{(rp)\oplus q}(\alpha)\right) \\
&= 2^{-|r|}M^{(rp)\oplus q}(x).
\end{aligned}$$

Applying the definition (8),

$$MI^q(rp:x) - MI^q(p:x) = \log_2\frac{M^{(rp)\oplus q}(x)}{M^{p\oplus q}(x)} \overset{+}{\leq} |r|.$$ □

A.2 Anytime prediction

In (4), p can be viewed as the code for an anytime algorithm that makes predictions according to successive approximations of μ_p. Indeed, we can simulate $U^\alpha(p)$ for all possible $\alpha \in \mathbb{B}^\infty$ in parallel: each time U reads a new bit of α, fork the computation into cases where the bit is `0` or `1`, allocating an equal share of time to each. Each time U outputs a bit, add 2^{-n} to our estimate of $\mu_p(x)$, where n is the number of bits read from α so far, and x is the output so far. Since these estimates are monotonically increasing and converge to $\mu_p(x)$, we say the prior μ_p is *lower-semicomputable* (if p is a finite program padded with **0**; otherwise, it is lower-semicomputable *relative to* p).

To complete the anytime prediction algorithm, we can prepend to p a constant instruction along with an encoding of the history $h \in \mathbb{B}^*$, to make it output successive estimates of the posterior

$$\mu_p(x \mid h) := \frac{\mu_p(hx)}{\mu_p(h)}.$$

A ratio of increasing estimates is not itself necessarily increasing, so the posterior is not in general lower-semicomputable. It is, however, *limit-computable* [LH15b, Section 3; Ste26, Section 3.3]: the ratio of estimates converges (non-monotonically) to the posterior $\mu_p(x \mid h)$.

Since our posterior predictions are only limit-computable, one might wonder about the performance of the broader class of anytime algorithms with this property. That is, instead of the lower-semicomputable prior μ_p, suppose we start with a general limit-computable prior; then the posterior, being a ratio of priors, would still be limit-computable.

One such algorithm may estimate the posteriors of M^q with sufficient precision to predict Putnam's adversarial x (of any desired length, see Section 4.1), which M^q fails to predict. While x (and hence, the point mass prior concentrated on it) is limit-computable, it costs an extreme amount of time, even compared to some useful approximations of M^q. For a concrete comparison, note that we can approximate M^q by simulating all possible programs p as above, allocating a fraction 2^{-n} of our computation time to each program prefix of length n. This approximation is competitive against every fast and short expert, with a runtime overhead "only" exponential in the expert's length. In contrast, to approximate M^q well enough to produce an adversarial x, we must wait for even the slowest programs of the desired length to halt. The runtime is much worse than exponential; it grows faster than any computable function.

In general, given an anytime algorithm p and a time limit t, let $\nu_{p,t}$ denote the prior semimeasure that results from running p for t time steps. Since $\nu_{p,t}$ is computable, there exists some $r := r(t)$ such that $\mu_{rp} = \nu_{p,t}$: r encodes the time limit, along with some wrapper code to run the subsequent algorithm p until the specified time t, and normalize its output into a semimeasure. Applying Theorems 3 and 6 to μ_{rp} yields the regret bound

$$L(M^q, x) - L(\nu_{p,t}, x) \overset{+}{\leq} MI^q(rp : x) \overset{+}{\leq} |r| + MI^q(p : x). \tag{9}$$

By setting t to the kth prefix-free busy beaver number [BS12], we can get $|r| \overset{+}{=} k$. As the busy beaver numbers grow faster than any computable function, an anytime algorithm can only substantially increase the right-hand side of (9) after an unfathomable amount of time.[9] This increase is more than a proof artifact: the sheer scale of busy beaver numbers makes them informative about some otherwise undecidable problems [Cha87].

B Additional considerations regarding oracle Solomonoff induction

B.1 The reference universal computer

Fundamentally, the information vantage point consists of not only the oracle q, but the entire relativized machine U^q. In the case of a finite prefix $q_{<n}$ padded with $\mathbf{0} := 0^\infty$, the oracle $q := q_{<n}\mathbf{0}$ can be hardcoded into the description of another universal computer V, such that $V^{\mathbf{0}} = U^q$. Our "oracle Solomonoff induction" is then just classical Solomonoff induction with respect to V.

Nonetheless, it is useful to separate out a "minimal" U to treat as a universal reference. The intuitive idea is as follows: while each person on Earth likely has a different vantage point, they are also likely to share some common knowledge. Provided that everyone can agree on a "Schelling point" machine U, we retain the freedom to write programs in our preferred languages, which translate onto U via interpreters that we include in our personal instantiation of q. U might be chosen to be accessible

[9] We are interested in times t at which $\nu_{p,t}$, or at least $L(\nu_{p,t}, x)$, has permanently settled to good enough approximations of its limit as $t \to \infty$; hence, no generality is lost by increasing t to the next prefix-free busy beaver number. These numbers are defined relative to a particular Turing machine, but another self-contained way to specify a similarly large time limit would be to describe a particular Turing machine that runs for a large number of steps before halting. (The program rp would alternate between simulating steps of this machine and steps of p, until the machine halts.) The longest-running 5-state machine takes 47,176,870 steps before halting [bbc+26]. With 6 states, a machine is known that runs for over $2 \uparrow\uparrow\uparrow 5$ steps [mxd25], the kind of number for which "the number of atoms in the Universe times the number of femtoseconds since the Big Bang" is a rounding error. An arbitrary k-state Turing machine takes $O(k \log k)$ bits to specify, giving $|r| = O(k \log k)$.

from physics [Deu13; Özk15; JW18; KW20; Wol24; EH25], and/or tailored to minimize the hidden constants in our regret bounds [LV19, Section 3.9]. By agreeing on U, we ensure that the hidden constants in our results have small universal bounds independent of q, p, x. The oracle q is itself not a universal constant, so we explicitly account for its influence in our bounds.

What if multiple machines are common knowledge, say U, V, and W? Given U, there exists a constant instruction r, such that $U^{\bar{n}_1 \dots \bar{n}_k q}(\bar{r}ip) := T_{n_i}^{\bar{n}_1 \dots \bar{n}_k q}(p)$ for all $i, k, n_1, \dots, n_k \in \mathbb{N}$ and $p, q \in \mathbb{B}^\infty$ with $i \leq k$. Now set $\bar{n}_1, \bar{n}_2, \bar{n}_3$ to interpreters of U, V, W in the language of U, meaning that $T_{n_1} = U$, $T_{n_2} = V$, $T_{n_3} = W$. Then, $M^{\bar{n}_1 \bar{n}_2 \bar{n}_3 q}$, which is based on U, dominates the analogous semimeasures for the machines V, W, up to a constant factor that depends only on $|r|$. This becomes another universal constant that can be made small when selecting U; as long as we do this, the specific choice of U matters little.

B.2 Fresh restarting $M^{h\mathbf{0}}(x)$ vs lifelong learning $M^{\mathbf{0}}(hx)$

Suppose we want to predict $x \in \mathbb{B}^*$, using some prior history $h \in \mathbb{B}^*$. We can take either (1) a *fresh restarting* approach, directly predicting x according to the semimeasure $M^{h\mathbf{0}}(x)$ that accesses h via the oracle; or (2) a *lifelong learning* approach, predicting the concatenation hx according to the semimeasure $M^{\mathbf{0}}(hx)$, so that the predictor sees h before x.

According to Theorem 3, these two approaches optimize different objectives against different classes of experts. While both can access h while predicting x, a lifelong learner with sufficient experience must converge to the conclusion supported by h [Sol78], whereas a freshly restarted learner must give substantial weight to other alternatives.

For example, suppose $h := \mathtt{0}^n$ for some large n, while the first bit of x is $x_1 := \mathtt{1}$. In order to achieve bounded regret against a straightforward "always 0" expert, our predictions must converge as h gets longer; that is, $\lim_{n\to\infty} M^{\mathbf{0}}(\mathtt{1} \mid \mathtt{0}^n) = 0$. There is no need to hedge substantially against the possibility of seeing $x_1 = \mathtt{1}$, because we only compete against accessible experts; without knowing n, no expert that predicts the long string of 0s will know to change to 1 after exactly n steps. In contrast, the fresh restarting learner must compete with experts that are *only* evaluated on x, and must therefore hedge against both possibilities for x_1! Formally, $0 < \inf_{h\in\mathbb{B}^*} M^{h\mathbf{0}}(\mathtt{1}) < \sup_{h\in\mathbb{B}^*} M^{h\mathbf{0}}(\mathtt{1}) < 1$.

More generally, given a long sequence consisting of multiple episodes, predicting each new episode $e^{(i)} \in \mathbb{B}^*$ with a freshly instantiated universal semimeasure $M^{e^{(1)}\dots e^{(i-1)}\mathbf{0}}(e^{(i)})$ enables Theorem 3 to bound the regret separately per episode. The price is that the bound on *total* regret is now multiplied by the number of episodes. If we want to minimize the total loss across all episodes, then the lifelong learning Solomonoff induction $M^{\mathbf{0}}(e^{(1)} \dots e^{(N)}) = \prod_{i=1}^{N} M^{\mathbf{0}}(e^{(i)} \mid e^{(1)} \dots e^{(i-1)})$ is superior.

Prior work on Solomonoff induction generally focuses on the lifelong learning version, representing past episodes as history rather than a vantage point oracle. Its performance on future episodes can still be bounded, albeit with additional error terms [CHS07; RH11, Section 8.3].

B.3 Unknowability

In challenging settings involving out-of-distribution generalization, we hope to improve our learning algorithms. Since M^q gives an upper bound on the performance of all accessible predictors, estimating the performance of M^q indicates how much room there may be for improvement.

For example, image classifiers often use spurious features, e.g. distinguishing a water bird from a land bird by the presence of water in the background [Sag+20]. Is this behavior a failure of our learning algorithms, or is it genuinely the best inference supported by our training data? If M^q successfully avoids mistakes that our classifier makes on out-of-distribution data, then general-purpose algorithmic improvements may be possible. On the other hand, if M^q replicates the undesired behavior, then the flaw is not in the algorithm. Instead, we should carefully examine what assumptions or additional data lead us to intuit better inferences than those made by M^q, so that we can make the relevant information available to our learning algorithms.

Another application of reasoning about what M^q does not know is found in AI safety. Cohen and Hutter [CH20] and Ebtekar and Cohen [EC26] propose techniques to help prevent superintelligent agents from autonomously taking dangerous, novel courses of action, based on the inherent uncertainty that even maximally-intelligent Solomonoff induction must maintain about its environment.

B.4 Optimality in computable environments

In Theorem 3, the universal predictor's regret bound holds for every observation sequence x, but depends on how much information the rival predictor μ_p has about x. There is a dual bound, which does not depend on the rival predictor, but instead depends on the complexity of the observation-generating process, henceforth called the *environment*. If a computable probability measure μ_p generates x, then μ_p is also the Bayes-optimal predictor. Consequently, any mean regret bound that holds against μ_p will in fact apply against all rivals, computable or otherwise.

In this dual setting, observations are random variables, so optimality is measured with respect to the mean cross-entropy loss that a predictor ν accumulates in the first $t \in \mathbb{N}$ steps:

$$L_t(\nu,\, \mu) := \mathbb{E}_{x\sim\mu}\, L(\nu,\, x_{<t}) = -\,\mathbb{E}_{x\sim\mu} \log \nu(x_{<t}).$$

Corollary 7 (Optimality in computable environments)**.** *For all $p, q \in \mathbb{B}^\infty$ such that μ_p is a probability measure (i.e. not merely a semimeasure), and $t \in \mathbb{N}$, we have*

$$L_t(M^q,\, \mu_p) - \inf_\nu L_t(\nu,\, \mu_p) \overset{+}{\leq} KM^q(p).$$

Proof. Theorem 3 and Proposition 4 together imply

$$L(M^q,\, x) - L(\mu_p,\, x) \overset{+}{\leq} KM^q(p).$$

Taking x to be the first t bits generated by μ_p, and taking expectations,

$$L_t(M^q,\, \mu_p) - L_t(\mu_p,\, \mu_p) \overset{+}{\leq} KM^q(p).$$

Hence, for every semimeasure ν (computable or otherwise),

$$\begin{aligned} L_t(M^q,\, \mu_p) - L_t(\nu,\, \mu_p) &\overset{+}{\leq} KM^q(p) + L_t(\mu_p,\, \mu_p) - L_t(\nu,\, \mu_p) \\ &= KM^q(p) - \mathbb{E}_{x\sim\mu_p}\left[\log \frac{\mu_p(x_{<t})}{\nu(x_{<t})}\right] \\ &= KM^q(p) - D_{\mathrm{KL}}\left(\mu_p(\cdot_{<t}) \parallel \nu(\cdot_{<t})\right) \\ &\leq KM^q(p). \end{aligned}$$

In the last line, we used the fact that the Kullback-Leibler divergence is non-negative even when the second argument is a semimeasure; normalizing would only decrease its value. Taking the supremum of both sides over all semimeasures ν completes the proof. □

We can compare the pair of optimality results for oracle Solomonoff induction: Theorem 3 applies to all possible observations but has a slack equal to the information content of the rival, whereas Corollary 7 applies against all rivals but has a slack equal to the relativized complexity of the stochastic process generating the observations.

The latter framing is more common in learning theory, where one often starts with probabilistic assumptions and then seeks an optimal learner. For example, a statistical mechanics perspective on learning assumes that the data is sampled from a "teacher" model, which is itself sampled from some prior, and then the student's prediction loss is averaged over the sampling of both the teacher and the data [SST92; WRB93; ZK16; Aub+19]. The computable world model of Corollary 7 includes most such settings: if the data $x \sim \mu_p$ consists of concatenated self-delimiting tokens sampled i.i.d. from a teacher model, then $KM^q(p)$ is approximately the model's complexity, which is small for a wide variety of statistical models used in practice.

Despite its intuitive appeal, when it comes to general online learning, we find the framing of Corollary 7 to be weak in both philosophical and practical terms. The information vantage point cannot constrain the environment (as some of it is inaccessible until observed), so the relativized complexity $KM^q(p)$ may well be high. This weakness is well-known, and historically contributed to skepticism regarding the efficacy of Solomonoff induction [Ste16]. In contrast, the information vantage point *does* constrain which rival predictors are accessible to us; therefore, combining the physical Church-Turing thesis with Theorem 5 results in a very small slack $MI^q(p : x)$ for Theorem 3.

Another important difference is that Corollary 7 assigns "objective" probabilities to the observations x, effectively splitting their complexity into a "model part" given by the description of μ_p, and a

"noise part" as x is sampled from μ_p. While this seems natural in a lot of settings, it leaves some ambiguity: for example, if many generating distributions are initially plausible, should we take μ_p to be (1) their Bayesian mixture, (2) the specific distribution that best fits observations, or (3) a point mass of probability one on the actual observed sequence? To avoid such ambiguities in the problem specification, we prefer to leave the task of probabilistic modeling up to the predictor. Indeed, the only probabilities referenced in Theorem 3 are given by the predictors. Instead of being imposed externally, these probabilities emerge from inference calculations on the actual observation x, offering a universal subjective interpretation of probability [Gil00; RH11; HH16, Part IV]. In a sense, this is another view of the meta-NFL problem (Corollary 2): we cannot be sure that the world adheres to some structure, but we can understand our own predictive capabilities.

B.5 Other loss functions

An important limitation of Theorem 3 is that it only applies to the log loss. The *strong convexity* of the log loss enables the predictor to hedge when it is uncertain [Vov01]. In contrast, many real settings require commitment to a specific action.

An important case is the 0-1 loss. Instead of probabilistic predictions, it demands a series of discrete predictions that the next bit x_t will be either `0` or `1`, and the loss is simply the number of mistakes. Given a Bayesian prior ν, it seems reasonable to choose the prediction that minimizes expected loss. For the 0-1 loss, this is the maximum a posteriori (MAP) estimate, so that

$$L_{\text{0-1}}(\nu, x) := \sum_{i=1}^{|x|} \mathbb{1}(\nu(x_{<i}x_i) \leq \nu(x_{<i}\neg x_i)).$$

Unfortunately, the adversarial example of Putnam [Put63] (also see Section 4.1) returns in full force to produce a sequence x on which this strategy – and in fact, *any* strategy – predicts every bit incorrectly. If the prior is M^q, no computable expert can predict this x; nonetheless, an always-`0` or always-`1` "expert" would guess correctly at least half of the time, resulting in a large regret. Since these experts are simple, Proposition 4 implies $MI^q(p : x) \stackrel{+}{=} 0$, so the bound in Theorem 3 does not hold.

On the other hand, this adversarial x does not seem like a "natural" sequence that could arise, for example, by sampling from a computable distribution. Its construction depends on the incomputable semimeasure M^q, which is related to the *Turing jump* q' of q [Nie09; DH10]. We speculate that it may be possible to bound the regret for general loss functions in terms of $MI^q(p \oplus q' : x)$, leaving the precise expression and analysis to future work.

C No Free Lunch and complexity measures

No Free Lunch means that any generalization bound we prove must depend on an inductive bias. How can we understand the inductive bias implicit in some classical statistical learning results?

C.1 Uniform convergence

We will first review some relevant textbook results. More details are available from many sources, such as the recent book by Bach [Bac24, Section 4.5].

Consider a learning problem where we observe independent data $Z = (z_1, \ldots, z_m) \sim \mathcal{D}^m$ – i.e. the z_i are i.i.d. from $\mathcal{D}$ – and wish to identify a hypothesis $h \in \mathcal{H}$ having small risk $L_{\mathcal{D}}(h) = \mathbb{E}_{z\sim\mathcal{D}}\,\ell(h, z)$, often based on the empirical risk $L_Z(h) = \frac{1}{m}\sum_{i=1}^m \ell(h, z_i)$. In typical prediction settings, we have $z = (x, y)$ and $\ell(h, (x, y))$ might be $\mathbb{1}(h(x) \neq y)$, $\|h(x) - y\|^2$, or $-\log(h(x) \cdot y)$.

A primary tool of statistical learning theory is *uniform convergence*, based on the trivial inequality

$$L_{\mathcal{D}}(\hat{h}) \leq L_Z(\hat{h}) + \sup_{h\in\mathcal{H}}[L_{\mathcal{D}}(h) - L_Z(h)]. \tag{10}$$

If we have a reasonable upper bound on $\sup_{h\in\mathcal{H}}[L_{\mathcal{D}}(h) - L_Z(h)]$, then "what you see is what you get": predictors h with low training loss $L_Z(h)$ will actually generalize well (have small $L_{\mathcal{D}}(h)$).

For an empirical risk minimizer $\hat{h} \in \operatorname{argmin}_{h\in\mathcal{H}} L_Z(h)$, we further have for all $h^* \in \mathcal{H}$ that

$$L_Z(\hat{h}) \leq L_Z(h^*) = L_{\mathcal{D}}(h^*) + (L_Z(h^*) - L_{\mathcal{D}}(h^*))\,.$$

For any fixed h^*, we can apply a simple concentration inequality[10] (such as Hoeffding's) to show that $L_Z(h^*) - L_{\mathcal{D}}(h^*)$ is small. Then, if we choose that fixed h^* to have loss (arbitrarily close to) $\inf_{h\in\mathcal{H}} L_{\mathcal{D}}(h)$, we know $L_Z(\hat{h})$ will not be much bigger than $\inf_{h\in\mathcal{H}} L_{\mathcal{D}}(h)$. If we also have uniform convergence, meaning the last term in (10) is known to be small, then $\hat{h}$ will be nearly as good as the optimal predictor from $\mathcal{H}$.

It is worth noting that uniform convergence (10) gives only an upper bound on the generalization performance. While uniform convergence occurs iff the problem is uniformly learnable for binary classification [e.g. SB14, Theorem 6.7], the same is not true in more general settings. There are explicit constructions, for example, in multiclass classification [Nat89b, Section 6], mean estimation with missing data [SB14, Exercise 13.2], and high-dimensional linear classification [NK19, Section 3.1] and regression [ZSS20, Sections 3.1 and 3.2], where learning is possible but uniform convergence does not hold.[11]

C.2 Rademacher complexity

To show uniform convergence, we can use *Rademacher complexity*. Letting $\sigma_1, \dots, \sigma_m$ independently follow a Rademacher distribution $\mathrm{Uniform}(\{-1, +1\})$, the (expected) Rademacher complexity of a set of real-valued functions $\mathcal{F}$ is

$$\mathrm{Rad}_{\mathcal{D}^m}(\mathcal{F}) := \mathbb{E}_{z_1,\dots,z_m\sim\mathcal{D}}\,\mathbb{E}_{\sigma_1,\dots,\sigma_m}\left[\sup_{f\in\mathcal{F}}\frac{1}{m}\sum_{i=1}^{m}\sigma_i f(z_i)\right].$$

This definition arises from a technique called symmetrization, which yields [Bac24, Proposition 4.2]

$$\mathbb{E}_{Z\sim\mathcal{D}^m}\sup_{f\in\mathcal{F}}\left[\mathbb{E}_{z\sim\mathcal{D}}[f(z)] - \frac{1}{m}\sum_{i=1}^{m} f(z_i)\right] \le 2\,\mathrm{Rad}_{\mathcal{D}^m}(\mathcal{F}). \tag{11}$$

One can also use "desymmetrization" to show a closely related lower bound, although the form is slightly more complex [Kol06, Section 2.2; Wai19, Proposition 4.12].

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[10] Many sources instead bound $\sup_{h\in\mathcal{H}}|L_{\mathcal{D}}(h) - L_Z(h)|$, and so do not need separate control of $L_Z(h^*) - L_{\mathcal{D}}(h^*)$. While perhaps conceptually simpler, the description here makes clear that we only need to upper-bound $L_Z - L_{\mathcal{D}}$ for h^*, while we need to upper-bound $L_{\mathcal{D}} - L_Z$, or equivalently lower-bound L_Z, for all hypotheses. Thus asking for an upper bound everywhere is potentially wasteful. While with Rademacher complexity the gap in the bound is only a small constant, some methods [e.g. Men21] can give much tighter lower bounds than upper bounds. The hardness result of Nagarajan and Kolter [NK19], used to argue that "uniform convergence may be unable to explain generalization in deep learning," is also fundamentally about cases where $L_Z - L_{\mathcal{D}}$ is high; no such result is possible in the one-sided case [ZSS20, footnote 5].

[11] The examples of Natarajan [Nat89a, Theorem 6.7], Shalev-Shwartz and Ben-David [SB14, Exercise 13.2], and Zhou et al. [ZSS20, Section 3.1] apply to one-sided uniform convergence, as well as two-sided. While Zhou et al. left the one-sided failure as a conjecture, that conjecture is true when $d_S \ge 1$, as we now show.

The proof is identical to the two-sided case once we know $\mathbb{E}\,\lambda_{\max}(\Sigma - \hat{\Sigma})$ from their Proposition B.2 (which they denote ρ) is $\Omega(\sqrt{\lambda_n/n})$. To show this, fix a unit vector $u \in \mathbb{R}^{d_S}$ and let $q = -X_J^\mathsf{T} X_S u / \|X_J^\mathsf{T} X_S u\| \in \mathbb{R}^{d_J}$. Consider representing $\Sigma - \hat{\Sigma}$ in a basis beginning with the orthonormal vectors $(u, 0)$ and $(0, q)$; the upper-left 2×2 submatrix in that basis is

$$A := \begin{bmatrix} u^\mathsf{T}\left(I_{d_S} - \frac{1}{n}X_S^\mathsf{T}X_S\right)u & u^\mathsf{T}\left(-\frac{1}{n}X_S^\mathsf{T}X_J\right)q \\ q^\mathsf{T}\left(-\frac{1}{n}X_J^\mathsf{T}X_S\right)u & q^\mathsf{T}\left(\frac{\lambda_n}{d_J}I_{d_J} - \frac{1}{n}X_J^\mathsf{T}X_J\right)q \end{bmatrix} = \begin{bmatrix} 1 - \frac{1}{n}\|X_S u\|^2 & \frac{1}{n}\|X_J^\mathsf{T}X_S u\| \\ \frac{1}{n}\|X_J^\mathsf{T}X_S u\| & \frac{\lambda_n}{d_J} - \frac{1}{n}\|X_J q\|^2 \end{bmatrix}.$$

The largest eigenvalue of $\Sigma - \hat{\Sigma}$ is at least as large as $\lambda_{\max}(A)$. Let $Y = \|X_S u\|$. Since $X_J X_J^\mathsf{T} \to \lambda_n I$ as $d_J \to \infty$, we have $\|X_J^\mathsf{T}X_S u\|^2 \to \lambda_n Y^2$ and $\|X_J q\|^2 = \|X_J X_J^\mathsf{T} X_S u\|^2 / \|X_J^\mathsf{T} X_S u\|^2 \to \lambda_n$, with all convergences almost sure; thus $A \to M := \begin{bmatrix} 1 - \frac{1}{n}Y^2 & \frac{1}{n}\sqrt{\lambda_n}Y \\ \frac{1}{n}\sqrt{\lambda_n}Y & -\frac{1}{n}\lambda_n \end{bmatrix}$. As $\lambda_{\max}\left(\begin{bmatrix}\alpha & \beta \\ \beta & \gamma\end{bmatrix}\right) = \frac{1}{2}\left(\alpha + \gamma + \sqrt{(\alpha-\gamma)^2 + 4\beta^2}\right) \ge \frac{1}{2}(\alpha+\gamma) + |\beta|$, we have

$$\lambda_{\max}(\Sigma - \hat{\Sigma}) \ge \lambda_{\max}(A) \to \lambda_{\max}(M) \ge \frac{1}{2}\left(1 - \frac{1}{n}Y^2\right) - \frac{\lambda_n}{2n} + \frac{\sqrt{\lambda_n}}{n}Y.$$

Since u is a unit vector and X_S is standard normal, Y^2 is $\chi^2(n)$; hence $(1 - \frac{1}{n}Y^2)$ has mean zero. Y is $\chi(n)$, with mean $\sqrt{n-1} + O(1/n) = \sqrt{n} + O(1/\sqrt{n})$; thus $\mathbb{E}\,\sqrt{\lambda_n}Y/n = \sqrt{\frac{\lambda_n}{n}}\,(1 + O(1/n))$. Recalling that in the problem setup it was assumed $\lambda_n = o(n)$, we have shown that

$$\mathbb{E}\,\lambda_{\max}(M) \ge \sqrt{\frac{\lambda_n}{n}}\left(-\frac{1}{2}\sqrt{\frac{\lambda_n}{n}} + 1 + O\left(\frac{1}{n}\right)\right) = \Omega\left(\sqrt{\frac{\lambda_n}{n}}\right).$$

Following the proof of their Theorem 3.2, we thus have $\lim_{n\to\infty}\lim_{d_J\to\infty}\mathbb{E}\sup_{\|w\|\le\|\hat{w}_{\mathrm{MN}}\|}[L_{\mathcal{D}}(w) - L_S(w)] = \infty$.

Plugging in $\mathcal{F} = \{z \mapsto \ell(h, z) : h \in \mathcal{H}\}$ gives exactly a bound on $\mathbb{E}_{Z\sim\mathcal{D}^m} \sup_{h\in\mathcal{H}}[L_\mathcal{D}(h) - L_Z(h)]$. We can further obtain high-probability bounds, if ℓ is bounded, by McDiarmid's inequality [Bac24, Section 4.4.1].

Finally, we often bound the complexity of a loss class using the complexity of the hypothesis class. If $h(x) \in \mathbb{R}$ and $|\ell(h, (x, y)) - \ell(h', (x, y))| \le G|h(x) - h'(x)|$, then Talagrand's contraction lemma [Bac24, Proposition 4.3] implies that $\operatorname{Rad}(\mathcal{F}) \le G \operatorname{Rad}(\mathcal{H})$; thus

$$\mathbb{E}_{Z\sim\mathcal{D}^m} \sup_{h\in\mathcal{H}}[L_\mathcal{D}(h) - L_Z(h)] \le 2G \operatorname{Rad}(\mathcal{H}).$$

Let's turn for a moment to the problem of learning a linear function in some fixed feature space, $\mathcal{H}_\mathcal{W} = \{x \mapsto \langle w, \phi(x)\rangle : w \in \mathcal{W}\}$. Choosing $\phi(x) = x$ covers standard linear predictors. Choosing ϕ to map into some Hilbert space (here assumed to be over the reals for simplicity) corresponds to learning in a reproducing kernel Hilbert space (RKHS).

The usual bound for the Rademacher complexity of $\mathcal{H}_\mathcal{W}$ is

$$\begin{aligned}
\operatorname{Rad}_{\mathcal{D}^m}(\mathcal{H}_\mathcal{W}) &= \mathbb{E} \sup_{w\in\mathcal{W}} \frac{1}{m}\sum_{i=1}^m \sigma_i \langle w, \phi(x_i)\rangle \\
&= \mathbb{E} \sup_{w\in\mathcal{W}} \left\langle \frac{1}{m}\sum_{i=1}^m \sigma_i \phi(x_i), w \right\rangle \\
&\le \mathbb{E}\left\| \frac{1}{m}\sum_{i=1}^m \sigma_i \phi(x_i)\right\| \left(\sup_{w\in\mathcal{W}} \|w\|\right) \qquad \text{by Cauchy-Schwarz} \\
&\le \frac{1}{\sqrt{m}} \sqrt{\mathbb{E}_{x\sim\mathcal{D}}\|\phi(x)\|^2} \left(\sup_{w\in\mathcal{W}} \|w\|\right) \qquad \text{by Jensen and } \mathbb{E}[\sigma_i\sigma_j] = \mathbb{1}(i = j).
\end{aligned} \tag{12}$$

Since $\sup_{w\in\mathcal{W}}\|w\|$ appears so naturally here, it is tempting to think that it plays a vital role: low-norm functions are easy to learn, while high-norm functions are hard. This almost seems like a "built-in" inductive bias for learning functions from $\mathcal{H}_\mathcal{W}$.

This intuition is incorrect.

Proposition 8 ([Bac24, Exercise 4.9])**.** *Let $\mathcal{H}$ be any set of functions $\mathcal{X} \to \mathbb{R}$, $f : \mathcal{X} \to \mathbb{R}$ any function, and write $\mathcal{H} + \{f\} := \{x \mapsto h(x) + f(x) : h \in \mathcal{H}\}$. Then $\operatorname{Rad}_{\mathcal{D}^m}(\mathcal{H} + \{f\}) = \operatorname{Rad}_{\mathcal{D}^m}(\mathcal{H})$.*

Proof. Since f does not interact with h and $\mathbb{E}\,\sigma_i = 0$, we have

$$\mathbb{E}\left[\sup_{h'\in\mathcal{H}+\{f\}} \sum_i \sigma_i h'(x_i)\right] = \mathbb{E}\sup_{h\in\mathcal{H}}\left[\sum_i \sigma_i h(x_i) + \sum_i \sigma_i f(x_i)\right] = \mathbb{E}\sup_{h\in\mathcal{H}}\left[\sum_i \sigma_i h(x_i)\right]. \quad \square$$

In particular, translating the class is irrelevant: for any arbitrary w_0, we have $\operatorname{Rad}_{\mathcal{D}^m}(\mathcal{H}_\mathcal{W}) = \operatorname{Rad}_{\mathcal{D}^m}(\mathcal{H}_{\mathcal{W}-\{w_0\}})$. Thus,

$$\operatorname{Rad}_{\mathcal{D}^m}(\mathcal{H}_\mathcal{W}) = \inf_{w_0} \operatorname{Rad}_{\mathcal{D}^m}(\mathcal{H}_{\mathcal{W}-\{w_0\}}) \le \frac{1}{\sqrt{m}} \sqrt{\mathbb{E}\|\phi(x)\|^2} \left(\inf_{w_0} \sup_{w\in\mathcal{W}} \|w - w_0\|\right).$$

That is, it is not the maximum norm but the *radius* of $\mathcal{W}$ that matters. While our previous upper bound (12) did indeed prefer low-norm predictors to high-norm ones, that was an artifact of our proof. In fact, we could have subtracted w_0 in (12) in the first place, "naturally" yielding the radius.

This is exactly the picture of Figure 1: the Rademacher bound only cares about the size of each hypothesis class. It does *not* tell you which functions to prefer. By Proposition 8, this is not specific to linear classes; *any* function class can be translated by *any* function without modifying the Rademacher complexity.

C.3 Combinatorial dimensions for classification

For binary classifiers, where both upper and lower bounds for learnability are controlled by the VC dimension [VC68; SB14, Theorem 6.7], an exactly analogous result holds. There is no such thing as an "inherently simple" binary classifier; it only matters how flexible the hypothesis class is.

Proposition 9 ([MRT18, Exercise 3.25])**.** *Define the operation* $a \,\texttt{xor}\, b := \mathbb{1}(a \neq b)$ *for* $a, b \in \{0, 1\}$. *Write* $\mathcal{H} \,\texttt{xor}\, f := \{x \mapsto h(x) \,\texttt{xor}\, f(x) : h \in \mathcal{H}\}$. *Then,* $\mathrm{VCdim}(\mathcal{H} \,\texttt{xor}\, f) = \mathrm{VCdim}(\mathcal{H})$.

Proof. Let $h_1, \ldots, h_N \in \mathcal{H}$ be hypotheses achieving the vectors of labels $y_1, \ldots, y_N \in \{0, 1\}^{|X|}$ for the points in a set X. Write y for the vector of labels achieved by f on X. Then $h_1 \,\texttt{xor}\, f, \ldots, h_N \,\texttt{xor}\, f$ achieve the labels $y_1 \,\texttt{xor}\, y, \ldots, y_N \,\texttt{xor}\, y$, with xor elementwise. As xor is a bijection, the number of labelings in each set is the same; thus $\mathcal{H}$ shatters X iff $\mathcal{H} \,\texttt{xor}\, f$ does. □

The same is true for multiclass classification, where learnability is jointly characterized by the Natarajan and DS dimensions [Coh+26].

Proposition 10. *Let* $\mathcal{H}$ *be a class of functions from* $\mathcal{X}$ *to* $\mathcal{Y} \subseteq \mathbb{N}$. *For each* $x \in \mathcal{X}$, *let* π_x *be an arbitrary bijection on* $\mathcal{Y}$, *and let* $\pi \circ \mathcal{H} = \{x \mapsto \pi_x(h(x)) : h \in \mathcal{H}\}$. *Then the Natarajan dimensions [Nat89a] of* $\mathcal{H}$ *and* $\pi \circ \mathcal{H}$ *are equal; the same is true for the DS dimension [DS14].*

Proof. The proof proceeds exactly as that of Proposition 9, but using the concepts of Natarajan shattering or DS shattering instead of VC shattering. □

C.4 PAC-Bayes

Another major technique for proving generalization is PAC-Bayes bounds; Alquier [Alq24] gives a recent overview. In these bounds, there is a *prior* and a *posterior* distribution over predictors, although these need not be related by the rules of Bayesian inference. There are many forms of PAC-Bayes bounds, but in general, they bound (the mean of) the random variable $L_{\mathcal{D}}(h) - L_Z(h)$, for h sampled from the posterior distribution, in terms of the KL divergence of the posterior from the prior.

Such bounds also satisfy a similar invariance to the actual form of the hypotheses, as the KL divergence is invariant to any invertible transformation. If we transform the prior and posterior in the same way, we obtain the same bound on the expected generalization gap.

C.5 Stability

Probably the only other widely-used technique for proving generalization bounds is *algorithmic stability*; this can apply outside of settings with uniform convergence. For instance, the most common definition is uniform stability:

Definition 11 ([BE02; EEP05])**.** *When* $Z = (z_1, \ldots, z_m)$, *let* $Z_{\neg i}$ *denote* $(z_1, \ldots, z_{i-1}, z_{i+1}, \ldots, z_m)$. *A (possibly randomized) learning algorithm* $\mathcal{A}$ *is* $\beta(m)$-uniformly stable *if for all* $m \geq 1$,

$$\sup_{\substack{Z \in \mathcal{Z}^m, i \in [m] \\ z \in \mathcal{Z}}} |\mathbb{E}_{\mathcal{A}}\, \ell(\mathcal{A}(Z_{\neg i}), z) - \mathbb{E}_{\mathcal{A}}\, \ell(\mathcal{A}(Z), z)| \leq \beta(m).$$

We say an algorithm is uniformly stable *if* $\lim_{m \to \infty} \beta(m) = 0$.

Uniformly stable algorithms bear guarantees on $\mathbb{E}_{\mathcal{A}}\left[L_{\mathcal{D}}(\mathcal{A}(Z)) - L_Z(\mathcal{A}(Z))\right]$ in terms of $\beta(m)$ [BE02; EEP05; BKZ20]. Once again, nothing in this definition privileges any particular form of hypothesis; it only matters how stable the algorithm's output is to small changes in the training set. The same is true for various other notions of stability [Sha+10].

D Epistemic Boltzmann brains

Cosmology is the branch of physics that studies the large-scale structure of the Universe. One criterion for evaluating a proposed model of the Universe is whether it predicts an overabundance of so-called *Boltzmann brains*. These are observers that arise from chance fluctuations in a thermal background, instantiating a brief moment of subjective experience – complete with illusory memories of an orderly past – before dissipating back into equilibrium. In models where Boltzmann brains (BBs) vastly outnumber ordinary observers (OOs) like ourselves, it is argued that by simple counting, we must conclude that we are almost certainly BBs. Since this conclusion would completely undermine the scientific method, it is argued that we should reject such models [BT87; DeS+10; Car20].

In this section, we argue a contrary view: the problem of BBs is not specific to particular models of the cosmos, but is in fact a kind of No Free Lunch theorem. Even if our Universe contains no BBs at all, **without an inductive bias, we are forced to conclude that we are probably Boltzmann brains**. On the other hand, Hutter [Hut10] and Müller [Mül20; Mül26] show that with a universal inductive prior, even observers who know their Universe is full of BBs will view themselves as OOs, and make ordinary inferences. Hence, there is no reason to judge a model by its prevalence of BBs.

We make four versions of the argument, first from a position of total epistemic ignorance, adding more generous assumptions at each subsequent iteration.

D.1 Total epistemic ignorance

Ideally, we should be able to make inferences about our surrounding world without any prior assumptions. Through our biological senses, we take a sequence of observations, which we hope to extrapolate into predictions about future observations. Unfortunately, without an inductive bias, the No Free Lunch Theorem applies exactly as stated in Proposition 1. That is, no matter what observations we have accumulated so far, it remains equally plausible to receive any sequence of future observations. Thus, with overwhelmingly high subjective probability, we should expect any apparent structure in our past to dissolve immediately, just as for a BB.

D.2 Knowledge of dynamics

Now we add the assumption that we know the dynamical laws of the physical Universe, but start with no knowledge of its particular state. Wolpert et al. [WRS25] argue that concluding we are BBs from such an assumption would be circular and contradictory, because a BB cannot scientifically infer the laws of physics. Nonetheless, it should not *hurt* us to assume we are endowed with knowledge of the laws, as if by divine inspiration.

Without an inductive bias, we may take a maximum entropy prior (which we assume is well-defined) over the possible states of the Universe. If we do, then our subjectively held Bayesian belief is identical to the large-scale equilibrium distribution of the Universe, known as *heat death*. By the second law of thermodynamics, the time-evolution of our belief distribution remains at maximum entropy at all future times. Thus, even if the Universe is objectively quite orderly, our belief remains equivalent to a BB at heat death, unable to make useful inferences about the state of our surroundings. Before expanding on some details of this argument, we add an even more generous assumption.

D.3 Control of dynamics

Now we add the assumption that we can choose the dynamical laws at will, subject to obeying the second law of thermodynamics, i.e. that entropy cannot decrease. In other words, while the previous argument assumed we were divinely inspired by knowledge of physics, here we assume that we are divinely endowed with the ability to *control* physics as we see fit. We will see that, even granting such unrealistic powers, it remains impossible to make useful inferences about our surroundings.

At first, this may seem absurd. If we want to learn the state of our environment, surely we can just look at it and record its value to our memory; or failing that, we apply our physics-bending powers to *make* the environment match our beliefs! The issue is that the second law of thermodynamics does not play well with maximum entropy priors.

To illustrate, consider an ordered pair (m_t, e_t) representing the state of a memory and an environment variable, respectively, at time t. For simplicity, we model the states as discrete. In order to measure e_t, we might like to implement the assignment operation $m_{t+1} := e_t$, which can also be represented as a map

$$(m,\, e) \mapsto (e,\, e).$$

However, this map is irreversible, and decreases the entropy of the joint system. Nature forbids irreversible operations in isolation; in reality, such measurements are only possible by dumping the excess entropy into a third system that is known to have low entropy [Ben82]. One valid implementation first initializes the memory to zero by dissipating entropy into a heat bath, after which we can implement the reversible (i.e. injective) map

$$(0,\, e) \mapsto (e,\, e).$$

Unfortunately, a maximum entropy prior assigns equal probability to every possible joint state of the Universe. Since every transformation that obeys the second law necessarily preserves the maximum entropy, our belief is unchanged by such transformations. Consequently, there is no way to be confident that we have set the memory to zero, let alone infer the state of anything else.

D.4 Control of dynamics *and* initialized memory

It seems strange to imagine not knowing the state of our own memory, since by definition the memory should be used to record our beliefs. In this final version of the argument, we start with some memory of a fixed size, say 1 GB, containing known initial values, a large fraction of which are zeros. Our prior is maximum entropy on everything in the Universe except for this memory, resulting in a total entropy that is 1 GB less than the maximum possible.

In this case, we can indeed perform some measurements, after which our memory's state becomes correlated with the environment. However, by the second law of thermodynamics, our belief distribution will never be more than 1 GB away from heat death. Converted to physical units, this gap is less than 10^{-13} J K^{-1} [Fra05; EH25], so our observations can never assure us of large-scale structure sufficient to extract a macroscopically useful quantity of work. Normally, we make large-scale inferences by generalizing from smaller amounts of data; in contrast, the near-maximum entropy prior forces us to conclude the smallest possible correlation consistent with our measurements. Once again, we are effectively trapped as epistemic BBs.

As a final note, Wolpert et al. [WRS25] propose that we should believe a maximum entropy distribution on the trajectory of the Universe, conditional on two snapshots of its macrostate: the Big Bang, and the present day. We take the position that their procedure is unjustified, since there is no direct way to obtain those snapshots. Instead, scientific inferences about the Universe are more readily justified via (approximate) universal induction on real observations [Hut10; Mül20; Mül26].